

HAMHUNG, Korea, Democratic People's Republic ofWMO: **470410**Lat: **39.933N**Lon: **127.550E**Elev: **72**StdP: **14.66**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.9	12.2	-11.9	2.9	16.5	-7.3	3.7	18.5	21.2	19.2	19.9	18.6	4.1	50	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.3	89.9	76.0	86.8	74.8	84.0	73.3	79.1	86.0	77.4	83.7	75.8	81.3	7.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	141.9	83.3	75.5	133.7	81.1	74.2	127.9	79.5	42.8	86.2	41.0	84.1	39.5	81.2	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	15.3	12.2	3.2	94.1	4.0	2.6	0.4	96.0	-2.0	97.5	-4.2	99.0	-7.1	100.9		
(5)			1.5	81.6	3.5	2.4	-1.0	83.4	-3.1	84.8	-5.1	86.2	-7.6	87.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	25.9	30.3	39.9	51.5	61.1	68.2	74.3	75.0	66.7	56.0	42.1	30.6	(6)
(7)		DBStd	18.02	6.21	6.29	6.66	7.10	6.23	5.41	5.46	4.26	4.58	5.80	7.40	6.65	(7)
(8)		HDD50	2555	746	551	322	64	2	0	0	0	0	17	252	601	(8)
(9)		HDD65	5621	1211	971	779	408	153	28	2	1	33	282	688	1065	(9)
(10)		CDD50	3255	0	0	9	109	346	545	753	774	501	203	15	0	(10)
(11)		CDD65	846	0	0	0	3	33	123	291	310	84	2	0	0	(11)
(12)		CDH74	5808	0	0	0	59	402	837	2100	2011	386	13	0	0	(12)
(13)		CDH80	1693	0	0	0	8	100	251	731	542	61	0	0	0	(13)
(14)	Wind	WSAvg	3.6	4.0	4.0	4.2	4.2	4.1	3.3	2.9	2.7	3.0	3.4	3.5	4.1	(14)
(15)	Precipitation	PrecAvg	33.5	0.5	0.7	0.8	1.9	2.6	3.8	8.4	6.7	4.4	1.7	1.7	0.7	(15)
(16)		PrecMax	50.1	1.7	3.4	3.2	5.4	5.5	9.3	19.4	21.4	11.5	4.7	9.3	5.2	(16)
(17)		PrecMin	16.4	0.0	0.0	0.2	0.2	0.4	0.7	0.5	1.1	0.7	0.1	0.0	0.0	(17)
(18)		PrecStd	8.8	0.4	0.9	0.6	1.3	1.5	2.3	4.5	4.4	2.9	1.2	1.9	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	53.0	68.2	80.5	87.6	91.8	94.0	92.0	85.9	75.8	66.0	51.4	(19)
(20)			MCWB	37.6	43.0	53.7	59.3	65.9	69.9	79.0	78.9	72.6	62.6	53.8	42.5	(20)
(21)		2%	DB	41.9	48.5	61.2	74.7	82.6	86.2	90.6	88.6	81.3	72.0	61.0	47.3	(21)
(22)			MCWB	34.8	39.7	48.1	56.5	63.4	69.6	77.1	77.9	69.0	60.8	51.3	39.7	(22)
(23)		5%	DB	39.0	44.5	56.5	70.0	78.5	82.1	87.4	85.3	78.2	69.2	57.4	44.1	(23)
(24)			MCWB	33.0	36.8	45.1	54.6	61.9	68.9	76.3	76.6	68.3	59.1	48.7	37.9	(24)
(25)		10%	DB	36.2	40.9	52.1	65.8	74.5	78.7	84.3	82.7	75.6	66.4	53.9	41.1	(25)
(26)			MCWB	31.0	34.4	42.5	52.0	60.3	67.7	75.4	75.1	67.2	57.8	46.9	35.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.0	44.3	55.1	61.8	70.2	74.9	82.5	82.1	75.4	66.4	55.8	44.3	(27)
(28)			MCD B	44.5	50.9	65.3	73.8	81.1	83.9	89.8	88.5	83.0	71.8	62.3	48.6	(28)
(29)		2%	WB	35.9	40.4	49.5	58.5	66.4	72.1	79.4	79.4	72.7	63.1	52.8	41.1	(29)
(30)			MCD B	40.3	47.0	58.3	71.2	79.0	81.6	87.0	85.7	78.0	68.9	58.9	45.9	(30)
(31)		5%	WB	33.7	37.4	46.0	55.8	63.8	70.3	77.6	77.9	70.6	61.1	50.4	38.4	(31)
(32)			MCD B	37.8	43.4	54.9	67.6	74.8	78.5	84.8	83.6	75.2	66.6	56.0	43.1	(32)
(33)		10%	WB	31.8	34.9	43.2	53.2	61.7	68.8	76.0	76.3	68.7	59.1	47.9	36.0	(33)
(34)			MCD B	35.4	39.8	51.3	63.8	71.5	76.0	82.3	81.2	73.6	64.4	53.0	40.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.8	17.6	17.3	19.4	18.4	14.4	11.7	12.3	15.7	17.8	16.3	16.6	(35)
(36)			MCDBR	20.0	22.0	24.7	27.5	28.2	22.7	18.6	16.6	19.9	21.7	21.6	19.3	(36)
(37)			MCWBR	15.0	15.2	14.8	14.1	13.5	10.2	8.4	8.2	10.0	12.4	14.1	14.0	(37)
(38)		5% WB	MCD BR	18.3	20.0	22.9	25.1	23.6	18.1	15.9	14.5	15.6	18.3	18.9	17.8	(38)
(39)			MCWBR	14.4	14.5	15.1	14.6	13.1	9.4	8.3	8.0	9.4	12.0	13.2	13.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.382	0.450	0.527	0.602	0.626	0.675	0.620	0.547	0.486	0.490	0.438	0.382	(40)	
(41)		taud	2.021	1.883	1.753	1.656	1.663	1.635	1.811	1.987	2.067	1.934	1.987	2.057	(41)	
(42)		Ebn at Noon	233	232	227	221	220	209	220	233	239	217	211	221	(42)	
(43)		Edh at Noon	39	51	65	77	78	80	67	55	47	49	40	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	799	1084	1394	1601	1728	1580	1369	1358	1292	1080	758	689	(44)	
(45)		RadStd	54	78	113	134	174	157	185	134	135	88	104	58	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (18 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) +0.67	(i) +0.63	(j) N/A	(k) N/A	(l) +148
										(m) +76

Nomenclature: See separate page